Andrew A. Lane
Andrew.A.Lane@Stonybrook.edu

Education

Stony Brook University 2019–present

Ph.D. student, Electrical and Computer Engineering. Research focus: neuromorphic signal processing and EEG/ERP representation analysis. Advisor: Wendy Tang; committee/collaborator: Brady D. Nelson.

New York Institute of Technology 2010

M.S., Electrical and Computer Engineering.

Hofstra University 2008

B.S., Biomedical and Electrical Engineering.

Research Experience

Doctoral Researcher, Stony Brook University 2019–Present

Developed ARSPI-Net, a staged neuromorphic signal-processing framework for EEG/ERP analysis. Doctoral work models affective neural data as noisy, partially observed outputs of biological dynamical systems.

Bioinformatics Analyst, Feinstein Institute for Medical Research 2011–2014

Developed laboratory-automation robotics software and research data-management systems.

Research Analyst, Rockefeller University 2008–2011

Developed automation and robotics programming for high-throughput screening workflows.

Teaching and Academic Appointments

Adjunct Instructor, Hofstra University 2017–Present

Computer science and engineering instruction.

Adjunct Assistant Professor, Farmingdale State College 2014–Present

Computer science and engineering instruction.

Instructor, Stony Brook University 2022–2025

Department of Electrical Engineering.

Selected Publications

A. Lane, W. Tang, and B. Nelson, "Towards ARSPI-Net: Advancing EEG Feature Extraction with Neuromorphic Algorithms," *2024 IEEE Long Island Systems, Applications and Technology Conference (LISAT)*, pp. 1–8, 2024. doi:10.1109/LISAT63094.2024.10808138.

A. Lane, W. Tang, and B. Nelson, "Towards ARSPI-Net: Development of an Efficient Hybrid Deep Learning Framework," *2023 IEEE Long Island Systems, Applications and Technology Conference (LISAT)*, pp. 1–8, 2023. doi:10.1109/LISAT58403.2023.10179592.

Technical Skills

Neuromorphic computing; spiking neural networks; liquid-state machines; reservoir computing; graph neural networks; EEG/ERP signal processing; perturbation analysis; dimensionality reduction; Python; machine learning; LaTeX.